

How Many Sign Restrictions Are Needed? A Sharp Lower Bound for Identifying Multiple Shocks in SVARs*

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Abstract

This paper derives the exact minimum number of impact sign restrictions required to identify m shocks in a structural vector autoregression under a natural pairwise separation condition. The answer is $f(m) = (\lfloor \log_2 m \rfloor + 3)m - 2^{\lfloor \log_2 m \rfloor + 1}$. The proof follows from deriving a sharp lower bound using a Kraft-type inequality and then constructing an admissible restriction matrix that attains the bound. Our result reveals a significant sparsity leap: while traditional recursive orderings require $O(m^2)$ restrictions, sign-based separation requires only $O(m \log_2 m)$, providing a transparent benchmark for designing sparse identifying schemes in large SVARs.

Keywords: structural VAR, sign restriction, set identification, lower bound

JEL classifications: C11, C55, E50

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1 Introduction

Structural vector autoregressions (SVARs) remain a standard tool for empirical macroeconomics and finance since the seminal contribution of Sims (1980). In many modern applications, researchers prefer set-identifying restrictions based on signs or inequalities rather than exact recursive orderings or exclusion restrictions. The appeal of this approach is that it allows identification to be disciplined by economic theory while relaxing the controversial lead-lag hierarchies inherent in recursive orderings. By avoiding such unidirectional causal chains, sign-identified SVARs provide a framework more aligned with structural theory while remaining agnostic about impulse response magnitudes; see, among others, Faust (1998), Canova and Nicolo (2002), Uhlig (2005), and Amir-Ahmadi and Drautzburg (2021).

Before estimation, however, applied researchers face a basic design problem: how many impact restrictions are actually needed to keep the shocks of interest distinct? This question is conceptually separate from algorithm design and posterior inference. It is instead a combinatorial identification question about how sparse an admissible system of restrictions can be. In applied work, the answer matters because the number of restrictions affects transparency, the ease of economic interpretation, and, in large systems, computational feasibility. A sharp lower bound therefore provides a useful benchmark: any empirical design that uses fewer restrictions than this minimum cannot achieve pairwise shock separation and is therefore inadmissible for identifying the shocks.

To formalize the problem, let $\mathbf{y}_t = (y_{1,t}, \dots, y_{n,t})'$ follow a p -lag VAR:

$$\mathbf{y}_t = \mathbf{a}_0 + \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \quad (1)$$

$$\mathbf{u}_t = \mathbf{B}_0 \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \quad (2)$$

where the structural shocks $\mathbf{v}_t$ are linked to the reduced-form residuals $\mathbf{u}_t$ via the impact matrix $\mathbf{B}_0$, assumed to be non-singular. The covariance matrix of $\mathbf{u}_t$ is $\boldsymbol{\Sigma} = \mathbf{B}_0 \mathbf{B}_0' = \mathbf{B}_0 \mathbf{Q} \mathbf{Q}' \mathbf{B}_0' = \mathbf{B}_0^* \mathbf{B}_0^{*'} for any orthogonal matrix $\mathbf{Q}$, where $\mathbf{B}_0^* = \mathbf{B}_0 \mathbf{Q}$. Hence $\mathbf{B}_0$ is not (point-) identified. A standard set-identification strategy imposes sign restrictions on the structural parameters. Specifically, let $\mathbf{S}_0 = [s_{i,j,0}]$ be an $n \times m$ matrix collecting sign restrictions on impact (horizon $h = 0$) for m shocks of interest, where $s_{i,j,0} \in \{-1, 0, 1\}$. In a Bayesian framework (e.g., Rubio-Ramirez, Waggoner, and Zha (2010)), this entails$

drawing $\mathbf{Q}$ from a Haar uniform distribution on the orthogonal group $\mathbf{O}(n)$ and retaining only those structural matrices $\mathbf{B}_0\mathbf{Q}$ that are consistent with $\mathbf{S}_0$.

To ensure that the structural shocks in $\mathbf{S}_0$ are uniquely identifiable, the restriction matrix must satisfy the following pairwise separation condition (Chan, Matthes, and Yu, 2026): any two distinct shocks must agree in sign on at least one variable and disagree in sign on at least one other variable. This requirement ensures that no two shocks can be relabeled or sign-switched using impact information alone.

Assumption 1. For any $j \neq k$, $j, k = 1, \dots, m$, there exist i_1 and i_2 such that $s_{i_1,j,0} = s_{i_1,k,0} \neq 0$ and $s_{i_2,j,0} = -s_{i_2,k,0} \neq 0$.

This condition is consistent with standard empirical practice. For instance, in identifying structural demand and supply shocks (e.g., Uhlig, 2005; Peersman, 2005), a common strategy is to distinguish aggregate demand from supply innovations by their divergent impact on inflation, while they move output in the same direction. Without satisfying Assumption 1, these structural innovations would remain observationally equivalent, rendering their economic interpretation ambiguous.

Our main contribution is to derive the exact minimum number of non-zero impact sign restrictions compatible with Assumption 1. We find that the resulting formula is of order $O(m \log_2 m)$, which is substantially sparser than the $O(m^2)$ restrictions required by recursive orderings. The additional restrictions needed for each new shock grow only logarithmically, enabling sparse yet valid identification schemes in large SVARs. Methodologically, the proof follows a simple two-step logic. First, we establish a sharp lower bound by showing that the pairwise separation of shocks under sign restrictions satisfies a Kraft-type inequality—a standard result in information theory (see McMillan (1956) and Cover and Thomas (2006, Chapter 5))—to characterize the minimum requirements for identification. Second, we prove that this bound is attainable by constructing an explicit, recursive sign-restriction matrix.

The remainder of the paper is organized as follows. Section 2 characterizes the minimum number of sign restrictions through our main proposition. Section 3 provides the proof, establishing the result via a Kraft-type inequality and a recursive construction. Section 4 presents illustrative examples of optimal restriction matrices and concludes.

2 The Lower Bound

Let $f(m)$ denote the minimum possible number of non-zero entries in $\mathbf{S}_0$ over all matrices that satisfy Assumption 1. We now state the following proposition, which characterizes $f(m)$.

Proposition 1. *The minimum number of sign restrictions, denoted by $f(m)$, needed for an n -variable VAR to identify m shocks ($m \leq n$) is*

$$f(m) = (\lfloor \log_2 m \rfloor + 3)m - 2^{\lfloor \log_2 m \rfloor + 1}, \quad (3)$$

where $\lfloor \cdot \rfloor$ denotes the integer part of its argument.

From Proposition 1, we can see that while a Cholesky decomposition requires $m(m+1)/2 \approx O(m^2)$ restrictions to identify m shocks, our bound $f(m) \approx m \log_2 m$ shows that sign-based separation can be substantially more parsimonious in high-dimensional systems.

For example, when identifying $m = 15$ shocks, a traditional recursive ordering requires 120 restrictions, whereas our bound $f(15) = 74$ shows that nearly 38% fewer restrictions are needed for pairwise separation. This gap widens as m increases, illustrating the efficiency of sign-based identification in high-dimensional systems.

Table 1: Minimum Non-zero Entries: Cholesky vs. $f(m)$

#{Shocks} (m)	Cholesky ($m(m+1)/2$)	Our Bound ($f(m)$)	Sparsity Gain (%)
4	10	12	-20%
8	36	32	11%
15	120	74	38%
30	465	178	62%

Remark 1. *Our lower bound remains valid when ranking restrictions are incorporated. A ranking restriction, e.g., the response of variable i to shock j is at least as large as that of variable k to the same shock, can be transformed into a sign restriction by adding an auxiliary row (representing the ratio of the i -th variable over the k -th variable) to $\mathbf{S}_0$ with entries $s_{n+1,j,0} = 1$ and zeros elsewhere. Since the derivation of $f(m)$ does not depend on n (provided n is sufficiently large to accommodate such rows), the $O(m \log_2 m)$*

bound provides a valid floor for any identification scheme relying on these linear inequality structures.

3 Proof of Results

Write columns of $\mathbf{S}_0$ as $\mathbf{c}_1, \dots, \mathbf{c}_m \in \{-1, 0, 1\}^n$, and let

$$w_j = \#\{r : c_j(r) \neq 0\}$$

be the number of non-zero entries in column j , where $c_j(r)$ denotes the r -th element of $\mathbf{c}_j$. Then the object of interest is

$$f(m) = \min \sum_{j=1}^m w_j,$$

where the minimum is taken over all collections $\{\mathbf{c}_1, \dots, \mathbf{c}_m\}$ satisfying Assumption 1.

We define $|\mathcal{A}|$ as the cardinality of a set $\mathcal{A}$. $\emptyset$ is defined as the empty set. The proof has two parts. We first derive a sharp lower bound using a Kraft-type inequality. We then construct an admissible impact-sign matrix that attains this bound.

Proof of Proposition 1.

Part 1: a Kraft-type lower bound.

For each column $\mathbf{c}_j$, define the cylinder set

$$\mathcal{G}(\mathbf{c}_j) = \{\mathbf{x} \in \{-1, 1\}^n : x_r = c_j(r) \text{ whenever } c_j(r) \neq 0\}. \quad (4)$$

Since exactly w_j coordinates are fixed, the cardinality of this set is $|\mathcal{G}(\mathbf{c}_j)| = 2^{n-w_j}$. Likewise, $|\mathcal{G}(-\mathbf{c}_j)| = 2^{n-w_j}$. We claim that the $2m$ cylinder sets

$$\mathcal{G}(\mathbf{c}_1), \mathcal{G}(-\mathbf{c}_1), \dots, \mathcal{G}(\mathbf{c}_m), \mathcal{G}(-\mathbf{c}_m)$$

are pairwise disjoint. To see this, fix $i \neq j$.

If there existed $\mathbf{x} \in \mathcal{G}(\mathbf{c}_i) \cap \mathcal{G}(\mathbf{c}_j)$, then on every row where both $\mathbf{c}_i$ and $\mathbf{c}_j$ are non-zero,

the two columns would necessarily have the same sign, because both must agree with $\mathbf{x}$. But Assumption 1 requires at least one row where the two columns have opposite signs. Hence

$$\mathcal{G}(\mathbf{c}_i) \cap \mathcal{G}(\mathbf{c}_j) = \emptyset.$$

Similarly, if there existed $x \in \mathcal{G}(\mathbf{c}_i) \cap \mathcal{G}(-\mathbf{c}_j)$, then on every row where both columns are non-zero, the entries of $\mathbf{c}_i$ and $\mathbf{c}_j$ would necessarily have opposite signs. But Assumption 1 also requires at least one row where the two columns have the same non-zero sign. Hence

$$\mathcal{G}(\mathbf{c}_i) \cap \mathcal{G}(-\mathbf{c}_j) = \emptyset.$$

The same argument yields

$$\mathcal{G}(-\mathbf{c}_i) \cap \mathcal{G}(\mathbf{c}_j) = \emptyset \quad \text{and} \quad \mathcal{G}(-\mathbf{c}_i) \cap \mathcal{G}(-\mathbf{c}_j) = \emptyset.$$

Finally,

$$\mathcal{G}(\mathbf{c}_j) \cap \mathcal{G}(-\mathbf{c}_j) = \emptyset,$$

because a column used to identify a shock must have at least one non-zero entry.

Therefore, the $2m$ cylinder sets are pairwise disjoint subsets of $\{-1, 1\}^n$. Summing their cardinalities gives $\sum_{j=1}^m 2 \cdot 2^{n-w_j} \leq 2^n$, since $|\{-1, 1\}^n| = 2^n$. Equivalently, $\sum_{j=1}^m 2^{-(w_j-1)} \leq 1$. If we define $\ell_j = w_j - 1$, then $\sum_{j=1}^m 2^{-\ell_j} \leq 1$. This is exactly the Kraft inequality for binary prefix codes; see McMillan (1956) and Cover and Thomas (2006, Chapter 5).¹ Now our problem is the following optimization problem²:

$$\min \sum_{j=1}^m w_j, \quad \text{subject to} \quad \sum_{j=1}^m 2^{-(w_j-1)} \leq 1, \quad w_j \in \mathbb{Z}^+, \quad (5)$$

where $\mathbb{Z}^+$ denotes the set of all positive integers. We now state the following claim whose proof is included in the appendix:

¹Our identification problem is in essence the same as a problem in information theory, which is minimizing the total codeword length of a binary prefix code with m codewords.

²One should notice that the Kraft-type inequality is a necessary condition for Assumption 1; hence the solution to this constrained optimization problem provides a valid lower bound for the original identification problem. However, the two problems are not equivalent, so this bound is not necessarily sharp. We will later show by explicit construction that it is in fact sharp in our setting.

Claim 1. Let $m = 2^p + r$ with $p = \lfloor \log_2 m \rfloor$ and $0 \leq r < 2^p$. The solution to the minimization problem (5) satisfies $\sum_{j=1}^m w_j^* = m(p+1) + 2r$, where exactly $2^p - r$ of the w_j^* equal $p+1$ and the remaining $2r$ equal $p+2$.

Part 2: a construction that attains the bound.

We now construct an admissible matrix with exactly $f(m)$ non-zero entries. Define a recursive procedure **Build**($\mathcal{Z}$) that takes a nonempty set of column indices $\mathcal{Z}$ and returns a collection of rows:

- If $|\mathcal{Z}| = 1$, the **Build**($\mathcal{Z}$) procedure stops for $\mathcal{Z}$.
- Otherwise, split $\mathcal{Z}$ into two disjoint subsets $\mathcal{L}$ and $\mathcal{R}$ such that $|\mathcal{L}| = 2^{\lfloor \log_2 (|\mathcal{Z}|-1) \rfloor}$ and $\mathcal{L} \cup \mathcal{R} = \mathcal{Z}$, $\mathcal{L} \cap \mathcal{R} = \emptyset$. Add a row $\mathbf{R}_{\mathcal{Z}}$ where

$$R_{\mathcal{Z}}(j) = \begin{cases} 1 & \text{if } j \in \mathcal{L}, \\ -1 & \text{if } j \in \mathcal{R}, \\ 0 & \text{otherwise.} \end{cases}$$

Then recursively call **Build**($\mathcal{L}$) and **Build**($\mathcal{R}$) and take the union of all rows obtained.

Apply **Build** to the full set $\{1, \dots, m\}$. Finally, add an extra row $\mathbf{R}_0$ with all entries equal to 1.

We now state the second claim, whose proof is also in the appendix:

Claim 2. The above construction yields a matrix satisfying Assumption 1 with total number of sign restrictions exactly $f(m) = (\lfloor \log_2 m \rfloor + 3)m - 2^{\lfloor \log_2 m \rfloor + 1}$.

We have therefore constructed an admissible impact-sign matrix whose total number of non-zero entries equals the lower bound. The bound is sharp, and the claimed formula for $f(m)$ follows. Although one should notice that the admissible matrix which can achieve the lower bound is not unique.

4 Illustration and Concluding Remarks

4.1 Illustration

We illustrate the recursive construction from Step 2 in the previous section for $m = 4$ and $m = 5$. For $m = 4$, let $\mathcal{S} = \{1, 2, 3, 4\}$ be the set of column indices. Following the recursive procedure, we split $\mathcal{S}$ into $\mathcal{L} = \{1, 2\}$, $\mathcal{R} = \{3, 4\}$ and add row $\mathbf{R}_{\{1,2,3,4\}} = (1, 1, -1, -1)$. Recursively, $\text{Build}(\{1, 2\})$ splits into $\{1\}, \{2\}$; add row $\mathbf{R}_{\{1,2\}} = (1, -1, 0, 0)$. $\text{Build}(\{3, 4\})$ adds $\mathbf{R}_{\{3,4\}} = (0, 0, 1, -1)$. Finally add all-ones row $\mathbf{R}_0 = (1, 1, 1, 1)$. Then, one potential matrix that achieves the lower bound is

	1	2	3	4
$\mathbf{R}_{\{1,2,3,4\}}$	1	1	-1	-1
$\mathbf{R}_{\{1,2\}}$	1	-1	0	0
$\mathbf{R}_{\{3,4\}}$	0	0	1	-1
$\mathbf{R}_0$	1	1	1	1

Thus each column contains three non-zero entries, yielding a total of $4 \times 3 = 12$ restrictions, which equals $f(4)$.

For $m = 5$, we have $\mathcal{S} = \{1, 2, 3, 4, 5\}$, $|\mathcal{S}| = 5$. We split $\mathcal{S}$ into $\mathcal{L} = \{1, 2\}$ and $\mathcal{R} = \{3, 4, 5\}$ and add row $\mathbf{R}_{\{1,2,3,4,5\}} = (1, 1, -1, -1, -1)$. $\text{Build}(\{1, 2\})$ adds $\mathbf{R}_{\{1,2\}} = (1, -1, 0, 0, 0)$. $\text{Build}(\{3, 4, 5\})$: $|\mathcal{S}'| = 3$, $\lfloor \log_2(3 - 1) \rfloor = 1$, split into $\mathcal{L}' = \{3\}$ and $\mathcal{R}' = \{4, 5\}$; add $\mathbf{R}_{\{3,4,5\}} = (0, 0, 1, -1, -1)$. $\text{Build}(\{4, 5\})$ adds $\mathbf{R}_{\{4,5\}} = (0, 0, 0, 1, -1)$. Finally add all-ones row $\mathbf{R}_0 = (1, 1, 1, 1, 1)$. Then, one potential matrix that achieves the lower bound is

	1	2	3	4	5
$\mathbf{R}_{\{1,2,3,4,5\}}$	1	1	-1	-1	-1
$\mathbf{R}_{\{1,2\}}$	1	-1	0	0	0
$\mathbf{R}_{\{3,4,5\}}$	0	0	1	-1	-1
$\mathbf{R}_{\{4,5\}}$	0	0	0	1	-1
$\mathbf{R}_0$	1	1	1	1	1

Consequently, three columns have three non-zero entries each and two columns have four, giving a total of $2 \times 4 + 3 \times 3 = 17$ restrictions, which matches $f(5)$.

4.2 Concluding Remarks

This paper provides a sharp combinatorial benchmark for the design of sign-identified SVARs. Under a natural pairwise separation condition on impact sign restrictions, we derive the exact minimum number of non-zero restrictions needed to identify m shocks. In particular, the result shows that the total number of impact sign restrictions required for shock separation grows only on the order of $m \log m$, rather than quadratically in the number of shocks.

From an applied perspective, the result helps clarify what sparse identification can mean in large systems. Researchers often face a tradeoff between imposing enough restrictions to separate shocks and keeping the identifying scheme economically transparent. While too few restrictions fail to separate shocks, imposing an extensive set of constraints may lack solid theoretical backing and severely slow down posterior sampling. Proposition 1 characterizes the absolute minimum number of impact sign restrictions compatible with pairwise separation, and therefore provides a useful benchmark for evaluating different, and often richer, empirical identification schemes.

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Appendix A: Proof of Claims

In this appendix we provide a proof of the claims stated in the main text.

Proof of Claim 1. Let

$$K = \sum_{j=1}^m 2^{-(w_j-1)} \leq 1. \quad (\text{A.1})$$

Our objective is to characterize the sequence $\{w_j\}_{j=1}^m$ that minimizes the total sum $S = \sum_{j=1}^m w_j$ subject to (A.1). First, any optimal configuration must satisfy the constraint with maximum possible density. Suppose $K < 1$, and let $w_{\max} = \max_j \{w_j\}$. We observe that each term $2^{-(w_j-1)}$ in the Kraft sum can be expressed as a fraction with a common denominator $2^{w_{\max}-1}$, namely $2^{w_{\max}-w_j}/2^{w_{\max}-1}$. Thus, the total sum K can be written as $N/2^{w_{\max}-1}$ for some integer N . If $K < 1$, the integer N must be strictly less than $2^{w_{\max}-1}$, implying that the residual capacity $1 - K = (2^{w_{\max}-1} - N)/2^{w_{\max}-1}$ is at least $2^{-(w_{\max}-1)}$.

Consider now a perturbed set $\{w'_j\}$ where we decrease one instance of the largest weight by setting $w'_i = w_{\max} - 1$ for some i with $w_i = w_{\max}$, while keeping other weights unchanged. This modification increases the total Kraft sum by exactly $2^{-(w_{\max}-2)} - 2^{-(w_{\max}-1)} = 2^{-(w_{\max}-1)}$. Since this increment is no greater than the residual capacity $1 - K$, the new sum K' remains ≤ 1 , while the objective $S' = S - 1$ is strictly reduced. Thus, at the optimum, the gap $1 - K$ must be strictly less than $2^{-(w_{\max}-1)}$.

Second, we establish that the optimal weights must be quasi-uniform, differing by at most one. Suppose there exist w_a, w_b such that $w_a \geq w_b + 2$ for some $a, b \in \{1, \dots, m\}$. By shifting to a new set where w_a is replaced by $w_a - 1$ and w_b by $w_b + 1$, the objective S remains unchanged. However, the change in the Kraft sum is $\Delta K = 2^{-(w_a-2)} + 2^{-w_b} - (2^{-(w_a-1)} + 2^{-(w_b-1)}) = 2^{-(w_a-1)} - 2^{-w_b}$. Given $w_a - 1 \geq w_b + 1$, it follows that $\Delta K < 0$. This creates a strictly positive margin of at least $2^{-(w_a-1)}$, which, by the preceding density argument, allows for a further reduction of S . Therefore, any optimal distribution restricts w_j to the set $\{k, k+1\}$ for some $k \in \mathbb{Z}^+$.

To endogenously determine k , let n_1 denote the number of shocks with $w_j = k+1$. The remaining $m - n_1$ shocks have $w_j = k$. The requirement that (A.1) be binding implies:

$$n_1 2^{-k} + (m - n_1) 2^{-(k-1)} = 1, \quad (\text{A.2})$$

which yields $n_1 = 2m - 2^k$. Since $0 \leq n_1 < m$ must hold by definition, we have $2m - 2^k \geq 0$ and $2m - 2^k < m$. These inequalities imply $m \leq 2^{k-1} < 2m$, which uniquely identifies k as $\lceil \log_2 m \rceil$, or equivalently $\lfloor \log_2 m \rfloor + 1$ for non-power-of-two m . Substituting n_1 into the objective function, we obtain:

$$\begin{aligned} f(m) &= n_1(k+1) + (m - n_1)k = mk + n_1 \\ &= m(\lfloor \log_2 m \rfloor + 1) + 2m - 2^{\lfloor \log_2 m \rfloor + 1} \\ &= (\lfloor \log_2 m \rfloor + 3)m - 2^{\lfloor \log_2 m \rfloor + 1}. \end{aligned}$$

Note that if m is a power of two, n_1 reduces to zero, and the optimal configuration simplifies to $w_j = k$ for all $j = 1, \dots, m$, which remains consistent with the derived formula. $\square$

Proof of Claim 2. We verify that the recursive construction in Step 2 satisfies Assumption 1 and that the total number of non-zero entries equals $f(m)$.

Satisfaction of Assumption 1. Take any two distinct columns i and j . Consider the recursion tree implicit in the construction: at the top level, the set $\{1, \dots, m\}$ is split into $\mathcal{L}$ and $\mathcal{R}$ according to the rule $|\mathcal{L}| = 2^{\lfloor \log_2(m-1) \rfloor}$, $|\mathcal{R}| = m - |\mathcal{L}|$. The row added at this stage assigns 1 to columns in $\mathcal{L}$ and -1 to columns in $\mathcal{R}$. If i and j lie in different subsets at this top split, then this row already gives a pair of opposite signs. If they lie in the same subset, we move to the recursive call on that subset. This process continues until eventually a split separates them; the row added at that split gives opposite signs. Thus for every pair there exists a row with opposite signs. For a same-sign row, consider the split immediately above the first split that separates i and j (if such a higher split exists). At that higher split, i and j belong to the same subset, so the row added at that level assigns them the same non-zero sign. If i and j are separated already at the top split, then the all-ones row R_0 (which was added after all recursive calls) gives both columns the same sign 1. Hence every pair also has a row with the same non-zero sign. Therefore Assumption 1 holds.

Total number of non-zero entries. Let $f(m)$ denote the total number of non-zero entries for a system of size m . For any set $\mathcal{Z}$ with $|\mathcal{Z}| = m > 1$, the procedure adds one row $\mathbf{R}_{\mathcal{Z}}$ with m non-zero entries and recurses on two disjoint subsets $\mathcal{L}$ and $\mathcal{R}$. Including the

mandatory row $\mathbf{R}_0$, the total count satisfies the recurrence:

$$f(m) = f(k) + f(m - k) + m, \quad m > 1, \quad (\text{A.3})$$

where $f(1) = 1$ serves as the boundary condition and $k = 2^{\lfloor \log_2(m-1) \rfloor}$. We claim that for $m = 2^p + r$ with $0 \leq r < 2^p$ and $p = \lfloor \log_2 m \rfloor$, the solution is:

$$f(m) = m(p + 1) + 2r. \quad (\text{A.4})$$

We proceed by (strong) induction on m .

Base Case: For $m = 2$, we have $p = 1$ and $r = 0$. Equation (A.4) yields $f(2) = 2(1 + 1) + 2(0) = 4$, which matches the base case of the recurrence (A.3).

Inductive Step: Assume that for all system sizes $s < m$, the formula $f(s) = s(\lfloor \log_2 s \rfloor + 1) + 2(s - 2^{\lfloor \log_2 s \rfloor})$ holds. For a system of size m , the recursive construction (A.4) dictates:

$$f(m) = f(k) + f(m - k) + m, \quad \text{where } k = 2^{\lfloor \log_2(m-1) \rfloor}.$$

Let $m = 2^p + r$ with $0 \leq r < 2^p$. We consider two cases based on the remainder r :

Case 1: $0 \leq r < 2^{p-1}$. In this range, $m - 1 < 2^p + 2^{p-1} - 1$, which implies $k = 2^{\lfloor \log_2(m-1) \rfloor} = 2^{p-1} < m$. The system is partitioned into a left block of size $k = 2^{p-1}$ and a right block of size $m - k = 2^{p-1} + r < m$. Applying the inductive hypothesis:

$$\begin{aligned} f(2^{p-1}) &= 2^{p-1}((p-1) + 1) + 2(0) = p2^{p-1}, \\ f(2^{p-1} + r) &= (2^{p-1} + r)p + 2r. \end{aligned}$$

Substituting these into the recurrence relation:

$$\begin{aligned} f(m) &= f(2^{p-1}) + f(2^{p-1} + r) + (2^p + r) \\ &= p2^{p-1} + [p2^{p-1} + pr + 2r] + 2^p + r = m(p + 1) + 2r. \end{aligned}$$

Case 2: $2^{p-1} \leq r < 2^p$. In this range, $2^p \leq m - 1 < 2^{p+1} - 1$, which implies $k = 2^{\lfloor \log_2(m-1) \rfloor} = 2^p$. The system is partitioned into blocks of size $k = 2^p < m$ and $m - k =$

$r < m$. Applying the inductive hypothesis:

$$\begin{aligned} f(2^p) &= 2^p(p+1) + 2(0) = (p+1)2^p, \\ f(r) &= r(\lfloor \log_2 r \rfloor + 1) + 2(r - 2^{\lfloor \log_2 r \rfloor}) = rp + 2r - 2^p, \quad \text{since } \lfloor \log_2 r \rfloor = p-1. \end{aligned}$$

Substituting these into the recurrence relation:

$$\begin{aligned} f(m) &= f(2^p) + f(r) + (2^p + r) \\ &= (p+1)2^p + [rp + 2r - 2^p] + 2^p + r = m(p+1) + 2r. \end{aligned}$$

In both cases, the formula holds for m , completing the induction. Additionally, we note that for $m = 1$ ($p = 0, r = 0$), the formula yields $f(1) = 1(0+1) + 0 = 1$, which is also consistent. Finally, substituting $r = m - 2^p$ into the target formula yields:

$$f(m) = m(p+1) + 2(m - 2^p) = (p+3)m - 2^{p+1},$$

which matches the expression in Proposition 1.

Thus the constructed matrix is admissible and achieves the lower bound, proving that the bound is sharp.

□